SC4061 Computer Vision: Laboratory 1

Point Processing, Spatial Filtering, Frequency Filtering and Imaging Geometry

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1. Method and Verification

All seven experiments are implemented twice, and every number quoted in this report comes from an actual run rather than from expectation.

- **`Lab1.m`**: the complete source in one file, as required. It executes all seven sections end to end with no errors.
- **`lab1.py`**: an independent Python implementation used to render the figures and, more importantly, to act as a cross-check. Where the two implementations disagree, the disagreement is investigated and reported rather than averaged away.

The two agree throughout: identical Gaussian kernels, identical contrast limits, PSNR within 0.03 dB, an identical projective matrix, and an identical perceptron trace.

> **AMARAVEL BABU ARRAN's comment:** One section is deliberately exempt from that cross-check. GNU Octave's `histeq` is *not* numerically equivalent to MATLAB's. The Octave source carries an explicit `FIXME` saying so. For §2.2 I therefore implemented MATLAB's own documented `histeq` algorithm in the Python mirror instead of trusting Octave. The evidence that this was the right call is that the MATLAB algorithm lands at mean 127.502 against the theoretical ideal of 127.5, whereas Octave gives 128.823.

2.1. Contrast Stretching

(a) Reading the image

```
Pc = imread('mrt-train.jpg');
whos Pc            % 320x443x3 uint8  -> a colour image
P  = rgb2gray(Pc);
```

`whos Pc` reports **320×443×3 uint8**, exactly as the manual states.

> **AMARAVEL BABU ARRAN's comment:** I checked whether the colour was real. It is not: `max|R−G|` and `max|G−B|` are both **0**, so all three channels are identical and the file is a greyscale photograph merely *stored* in RGB. `rgb2gray` is therefore exact here rather than lossy, because the weights 0.299/0.587/0.114 sum to 1, so applied to three equal channels they return that same value. Worth knowing, because it means no information is discarded at this step. (Octave's reader collapses the file to 2-D on load, which is why `Lab1.m` guards the `rgb2gray` call with an `ndims` check.)

(c) Intensity range

```
min(P(:)), max(P(:))     % ->  13   204
```

min = 13, max = 204. The image uses only 192 of the 256 available levels, which is why it looks flat.

(d), (e) The stretch

Two lines, a subtraction followed by a multiplication, as the manual directs:

```
rmin = double(min(P(:)));  rmax = double(max(P(:)));
P2   = uint8(round((double(P) - rmin) * (255 / (rmax - rmin))));
```

Verification: `min(P2(:)) = 0`, `max(P2(:)) = 255`, exactly the required limits.

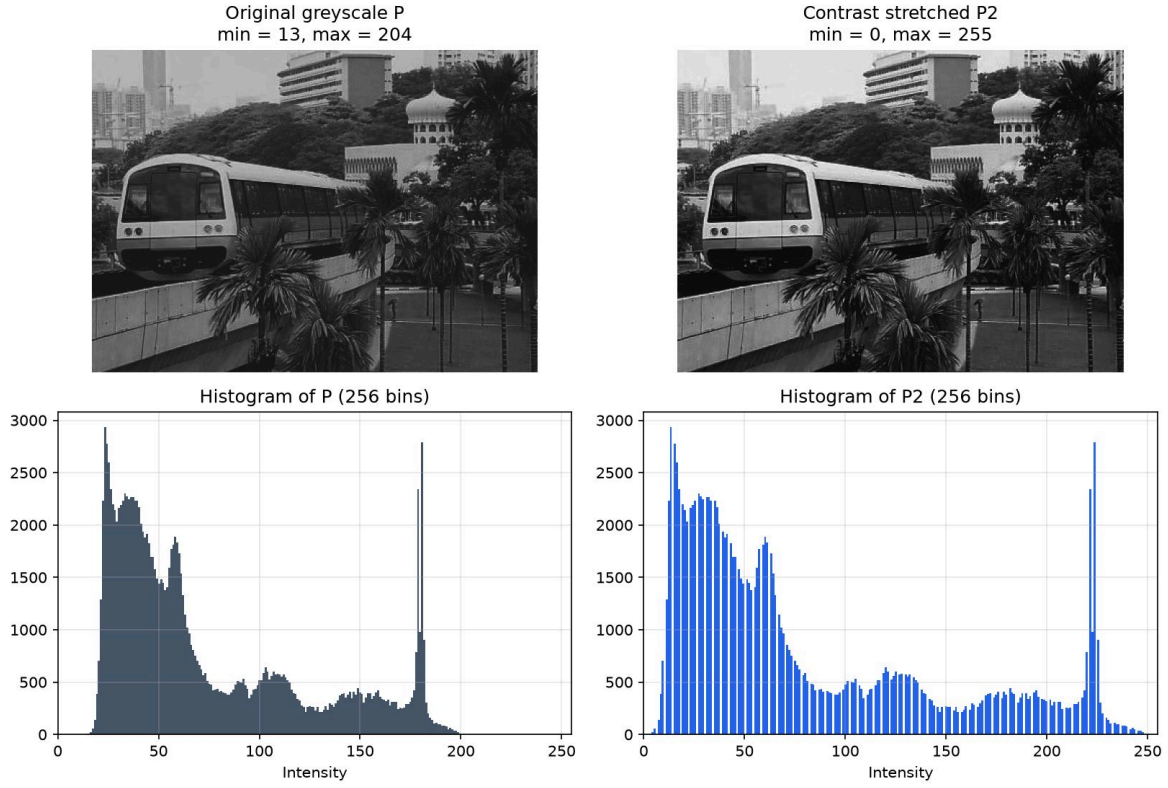


Figure 1. Contrast stretching of mrt-train.jpg. Top: the greyscale original, limited to levels 13 to 204, and the stretched result spanning the full 0 to 255. Bottom: the corresponding 256-bin histograms, showing that the transformation stretches the distribution without reshaping it.

> **AMARAVEL BABU ARRAN's comment:** The transformation is strictly linear, so the histogram is not reshaped, only stretched horizontally by 255/191 ≈ 1.335 and shifted to start at zero. The *shape* of the distribution, and therefore the relative tonal balance of the picture, is preserved exactly. This is the key difference from §2.2, and it is why contrast stretching never produces the unnatural look that equalisation can.

2.2. Histogram Equalization

(a) 10 bins versus 256 bins

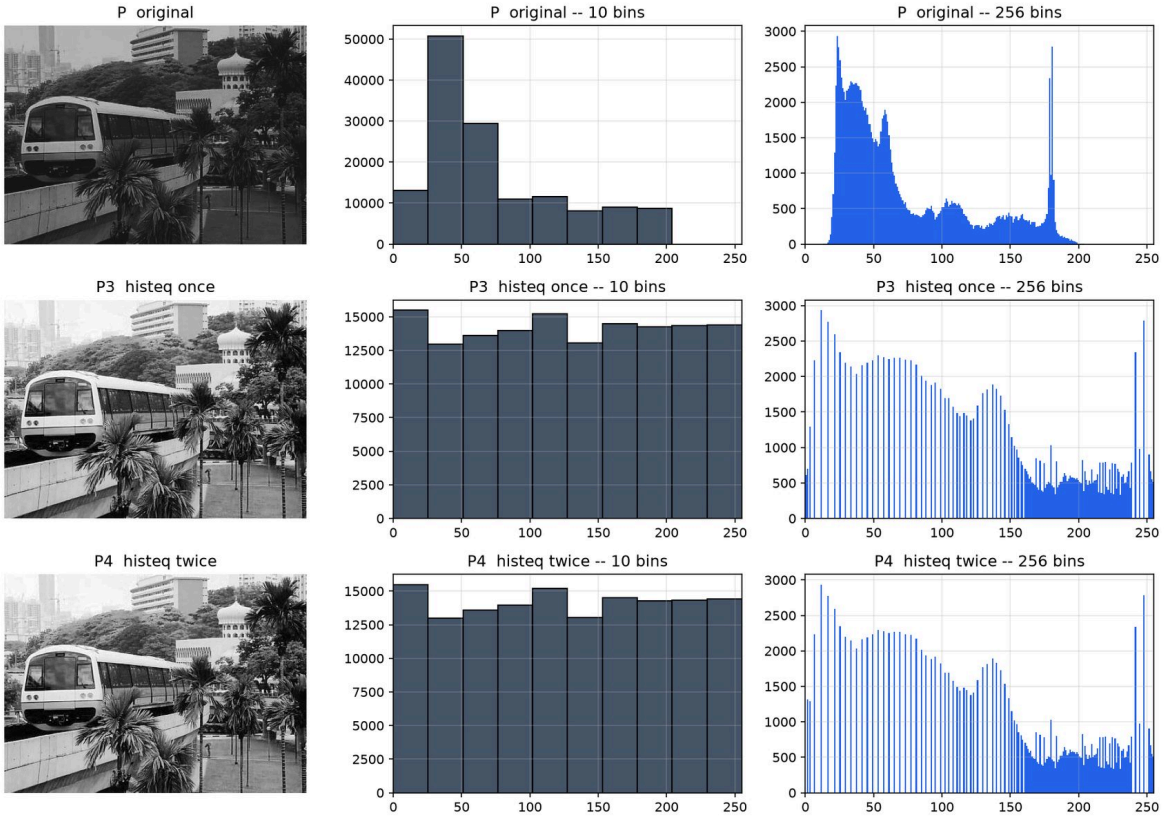


Figure 2. Histogram equalisation of mrt-train.jpg. Rows: the original, the result of one histeq pass, and the result of a second pass. Columns: the image, its 10-bin histogram and its 256-bin histogram. The 10-bin view appears flat while the 256-bin view reveals the comb of occupied and empty levels, and the second pass is visually indistinguishable from the first.

The 10-bin histogram is a coarse summary: each bar aggregates about 26 grey levels, so it shows only the broad shape: a large peak in the dark midtones and a long thin tail. The 256-bin histogram resolves every individual grey level and exposes structure the 10-bin view hides entirely, including a sharp isolated spike near level 180 and the hard cut-offs at 13 and 204.

(b) After equalization

```
P3 = histeq(P, 255);
```

Image	Mean	10-bin min/max count	Ratio
P original	73.989	0 / 50 769	50 769
P3 equalised once	**127.502**	12 965 / 15 488	**1.19**
P4 equalised twice	127.506	12 965 / 15 488	1.19

Are the histograms equalized? At 10 bins, essentially yes. The worst-to-best bin ratio falls from 50 769 to **1.19**, and the mean moves to 127.502, within 0.002 of the ideal midpoint 127.5.

Similarities and differences between the two views. Both describe the same image, and both show the distribution is now spread across the full range. But they give very different impressions of *how* equal it is. At 10 bins the result looks almost perfectly flat. At 256 bins it is visibly a **comb**: tall spikes separated by empty levels.

> **AMARAVEL BABU ARRAN's comment:** The comb is not a defect in the implementation; it is unavoidable. Histogram equalisation applies a single monotonic look-up table to the grey levels. Every pixel sharing an input level is mapped to the *same* output level, so a level containing 2 000 pixels cannot be spread across eight output levels to flatten the curve; it moves as one block. The algorithm can only *relocate* existing spikes, never split them. A flat 256-bin histogram is therefore impossible for any discrete image whose levels are unequally populated, and the 10-bin view only looks flat because binning hides the gaps.

(c) Re-running equalization

Running `histeq` a second time on `P3`:

- `max|P4 − P3| = 1`
- `mean|P4 − P3| = 0.0043`
- only **0.43 %** of pixels change at all, and those by a single grey level
- the 10-bin counts are **identical**

Does the histogram become more uniform? No. The operation is already at a fixed point.

> **AMARAVEL BABU ARRAN's comment:** The reason is that equalisation *is* the cumulative distribution function used as a transfer curve. Applying it once makes the CDF of the output approximately linear. Applying it again therefore maps through an approximately straight line, the identity. The only residual change is integer rounding on a handful of boundary pixels, which is exactly the 0.43 % observed. In exact arithmetic on a continuous distribution the second pass would be the identity precisely; the discrete case is as close to that as rounding permits.

2.3. Linear Spatial Filtering

(a) The Gaussian filters

```
function h = gaussianKernel(dim, sigma)
    r = (dim - 1) / 2;
    [x, y] = meshgrid(-r:r, -r:r);
    h = exp(-(x.^2 + y.^2) / (2 * sigma^2)) / (2 * pi * sigma^2);
    h = h / sum(h(:));            % normalise so the elements sum to 1.0
end
```

Both kernels sum to 1.000000.

	σ = 1.0	**σ = 2.0**
centre weight	0.162103	0.063191
corner weight	0.002969	0.023247
centre : corner	**54.6 : 1**	**2.7 : 1**

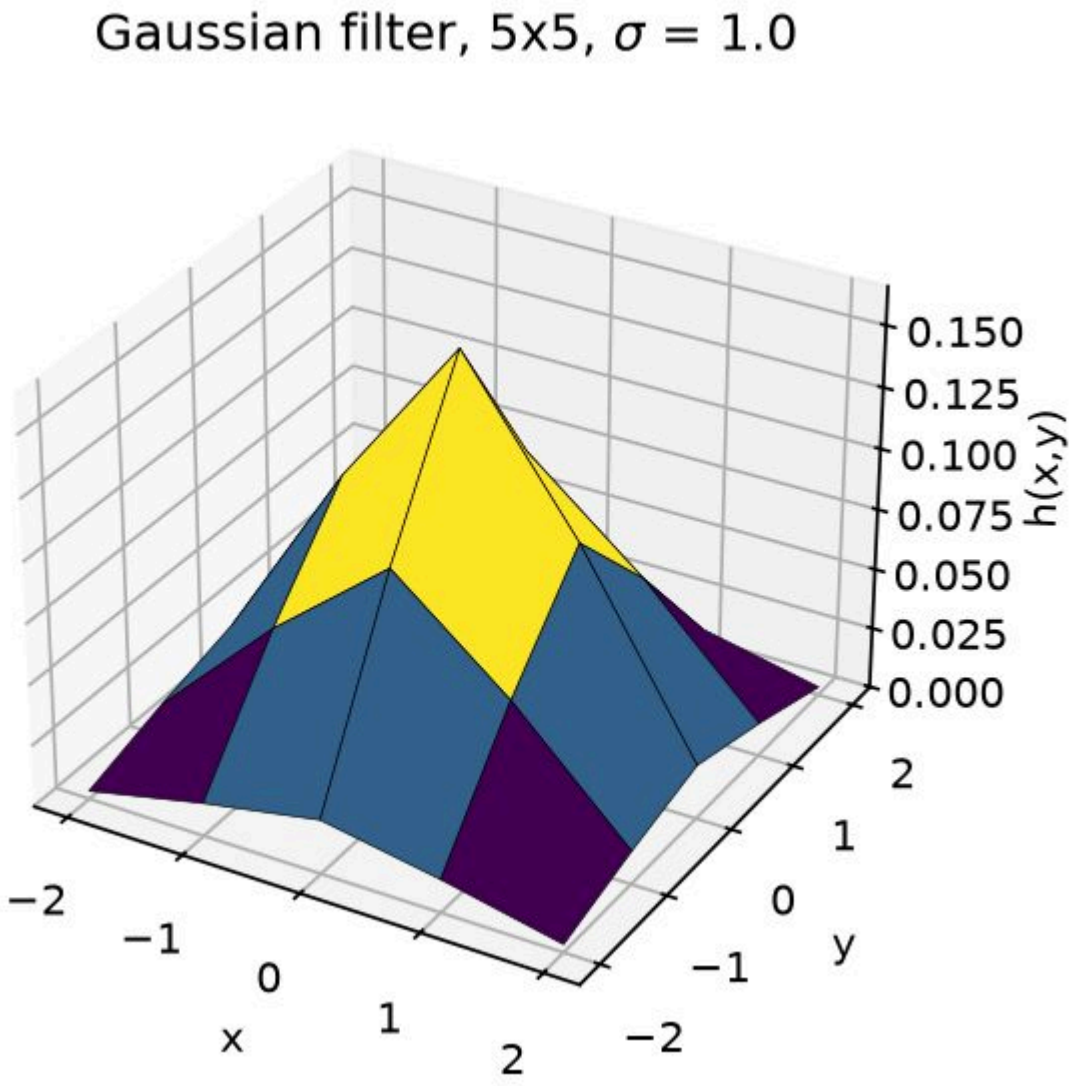


Figure 3. The 5x5 Gaussian averaging filter for sigma = 1.0, shown as a 3-D mesh. The kernel is sharply peaked, with a centre-to-corner weight ratio of 54.6 to 1.

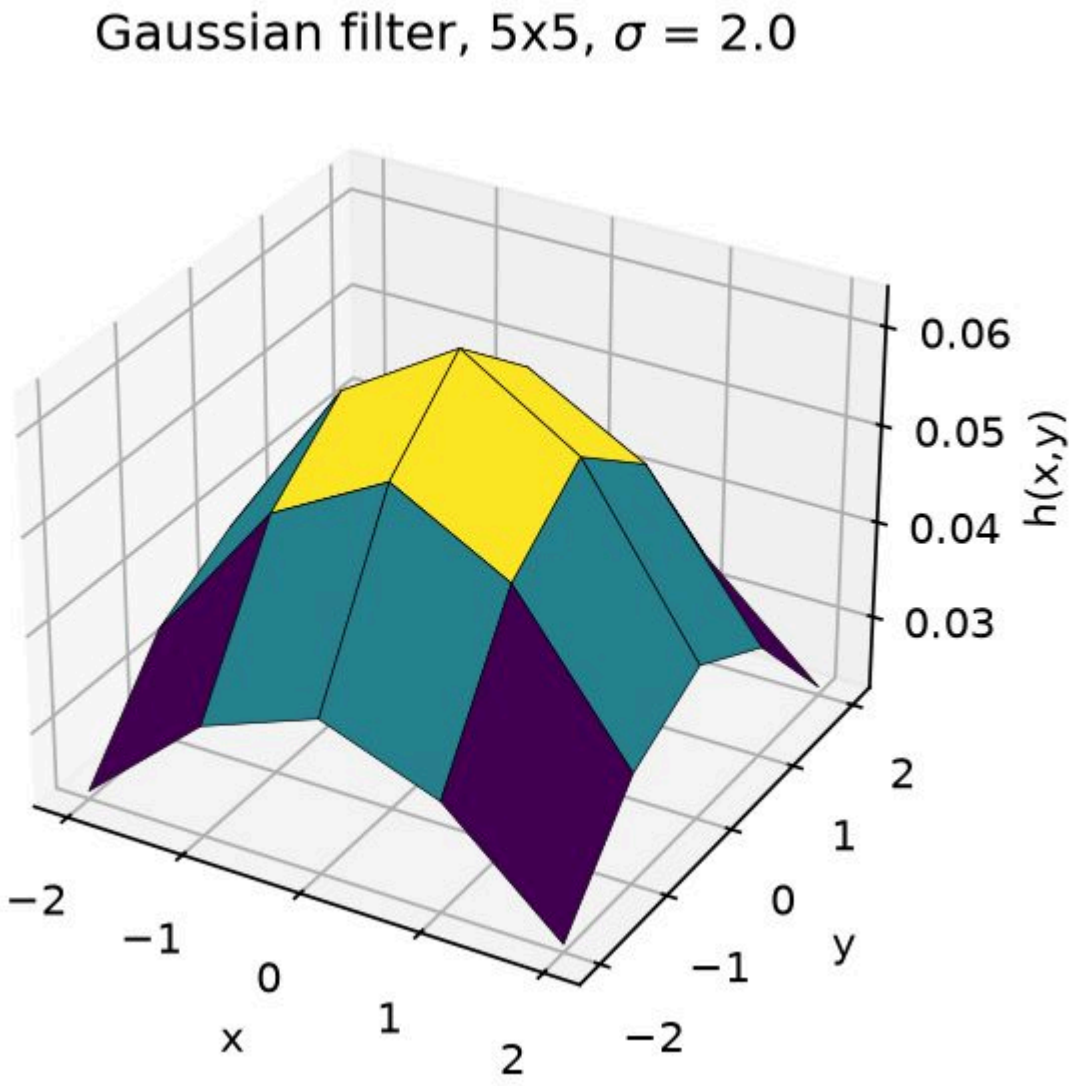


Figure 4. The 5x5 Gaussian averaging filter for sigma = 2.0. Truncation by the 5x5 window flattens the kernel to a centre-to-corner ratio of only 2.7 to 1, leaving it closer to a box average than to a true Gaussian.

> **AMARAVEL BABU ARRAN's comment:** The σ = 2.0 filter is badly truncated and this matters later. A 5×5 window reaches only ±2 pixels, which is ±1σ when σ = 2, so the kernel is cut off while still carrying substantial weight: its corner value is more than a third of its centre value. Renormalising to sum 1 hides the truncation but cannot undo it: the result is much closer to a plain 5×5 box average than to a true Gaussian. A faithful σ = 2 Gaussian needs a window of roughly 6σ ≈ 13 pixels. This is why σ = 2.0 never wins in any measurement below.

(b), (c) Gaussian noise: `lib-gn.jpg`

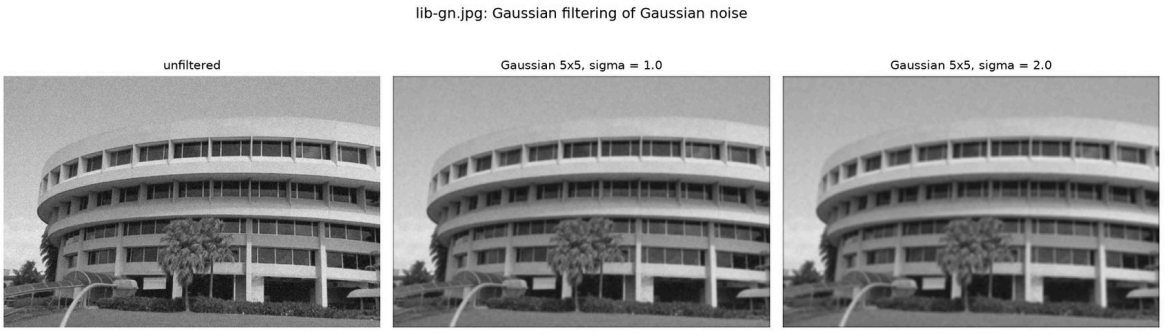


Figure 5. Gaussian filtering of lib-gn.jpg, which carries additive Gaussian noise. Left to right: unfiltered, sigma = 1.0 and sigma = 2.0. The grain is suppressed effectively, at a cost in edge sharpness that grows with sigma.

(d), (e) Speckle noise: `lib-sp.jpg`

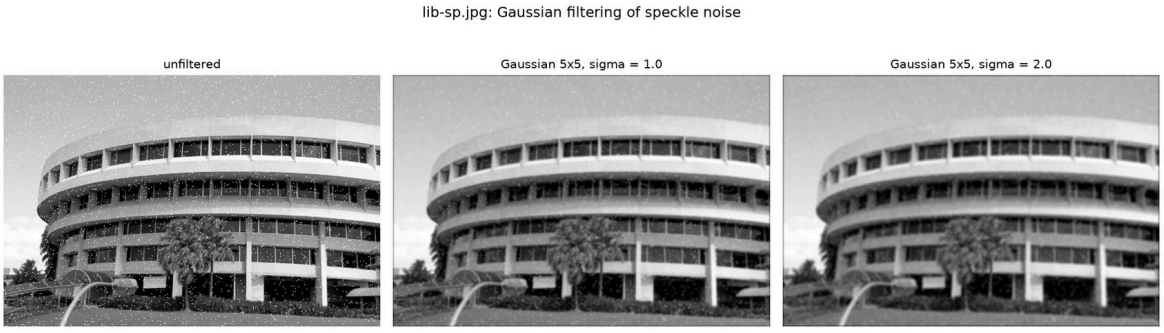


Figure 6. Gaussian filtering of lib-sp.jpg. The filters do not remove the impulses but spread each one across the kernel footprint, so the image stops looking speckled and begins to look smeared.

How effective are the filters? On additive Gaussian noise, clearly effective: the grain visibly disappears. On the speckle image the filters fail in a characteristic way: each bright and dark speck is not removed but *smeared* into a soft blob spanning the kernel footprint. The image stops looking speckled and starts looking dirty.

Trade-offs, including not filtering at all. Measured mean gradient magnitude, a proxy for retained edge sharpness:

	unfiltered	**σ = 1.0**	**σ = 2.0**
`lib-gn.jpg`	79.90	54.94 (−31 %)	49.30 (−38 %)
`lib-sp.jpg`	103.32	70.83 (−31 %)	62.06 (−40 %)

Not filtering keeps all detail and all noise. σ = 1.0 removes most of the visible grain for a 31 % loss of gradient energy. σ = 2.0 costs noticeably more sharpness for very little extra noise reduction, which the controlled experiment in §2.4 confirms is a bad bargain.

Are the filters better at Gaussian or speckle noise? Gaussian, decisively. The reason is visible in the pixel statistics: `lib-sp.jpg` has **1.41 % of pixels at exactly 0 and 2.39 % at exactly 255**.

> **AMARAVEL BABU ARRAN's comment:** That 3.8 % figure means the "speckle" image is really carrying **impulse (salt-and-pepper) noise**, not the multiplicative speckle the name suggests. This is the whole explanation for the filters' failure. A linear filter computes a weighted *mean*, and a mean has no resistance to outliers: a single 255 in a 5×5 window shifts the output of all 25 affected pixels. Averaging distributes the corruption instead of rejecting it. §2.4 shows the correct tool.

2.4. Median Filtering

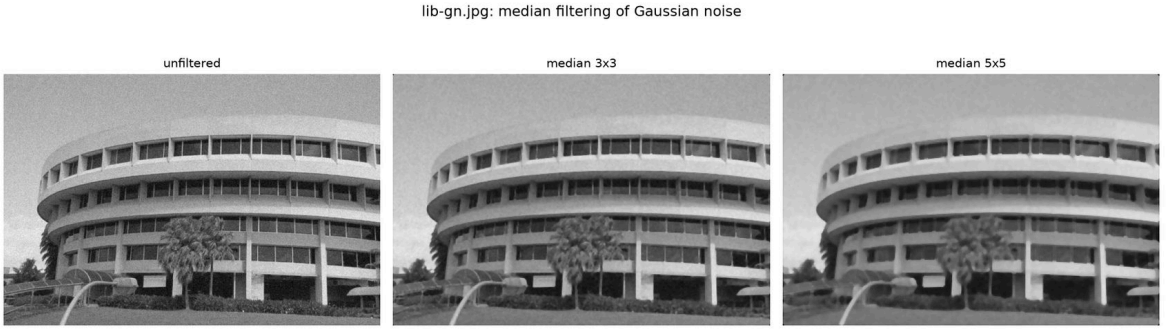


Figure 7. Median filtering of lib-gn.jpg at 3x3 and 5x5. The median is competitive with the Gaussian filter on additive noise but erases fine texture at the larger window size.



Figure 8. Median filtering of lib-sp.jpg. The impulses are rejected outright rather than spread, because a minority of extreme values cannot move a rank statistic.

(a) Making the comparison quantitative

Neither `lib-gn.jpg` nor `lib-sp.jpg` is supplied with a clean original, so no reference-based measure such as PSNR can be computed on them directly.

> **AMARAVEL BABU ARRAN's comment:** I first tried two reference-free substitutes and rejected both, because each is actively misleading here. Immerkaer's σ estimator assumes *uncorrelated* noise; a Gaussian low-pass leaves *correlated* residual noise, so the estimator reads far too low and would have "shown" Gaussian filtering beating median filtering by a factor of five. Counting pixels still pinned at 0 or 255 flatters convolution in the opposite direction, because a Gaussian filter does not remove an impulse it spreads it, so the pixel leaves the count while the damage remains. A measurement that cannot distinguish removing corruption from hiding it is worse than no measurement.

Instead I ran a **controlled experiment**: take a clean image, add noise of each type with known parameters, and measure PSNR against the known original.

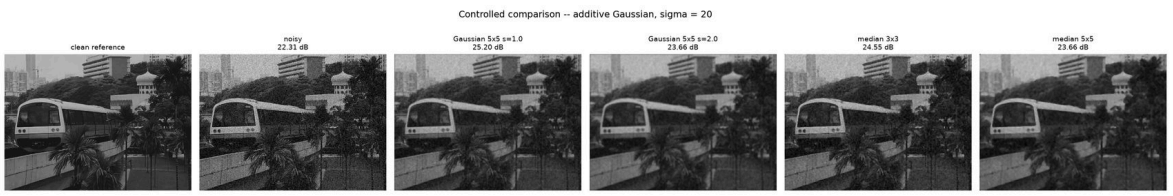


Figure 9. Controlled comparison on a clean reference corrupted by additive Gaussian noise of sigma = 20, with PSNR measured against the known original. The Gaussian filter at sigma = 1.0 is best at 25.20 dB.

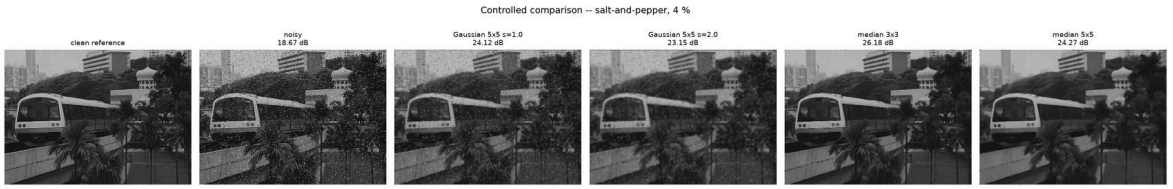


Figure 10. The same controlled comparison for 4 per cent salt-and-pepper noise. The 3x3 median leads at 26.18 dB, 2.06 dB ahead of the best Gaussian filter, reversing the ranking of the previous figure.

Filter	Additive Gaussian σ = 20	Salt-and-pepper 4 %
unfiltered	*22.31 dB*	*18.67 dB*
Gaussian 5×5, σ = 1.0	**25.20 dB** (+2.89)	24.12 dB (+5.45)
Gaussian 5×5, σ = 2.0	23.66 dB (+1.35)	23.15 dB (+4.48)
median 3×3	24.55 dB (+2.24)	**26.18 dB** (+7.52)
median 5×5	23.66 dB (+1.35)	24.27 dB (+5.61)

(b) How does Gaussian filtering compare with median filtering?

It depends entirely on the noise type, and the crossover is decisive.

- **Additive Gaussian noise**: the Gaussian filter wins (25.20 vs 24.55 dB). This is expected: for additive white Gaussian noise the optimal linear estimator *is* a weighted local mean, and the Gaussian kernel is close to it. The median is competitive but discards information by keeping only the rank statistic.
- **Salt-and-pepper noise**: the median wins by a wide margin, 26.18 dB against 24.12 dB, a **2.06 dB** advantage, and it is +7.52 dB over the unfiltered image versus +5.45 dB for the best Gaussian.

(c) Trade-offs

1. **Outlier rejection.** The median of a 3×3 window is unchanged by up to four corrupted neighbours; the mean is shifted by every one of them. This is the source of the entire salt-and-pepper result.
2. **Window size cuts both ways.** 5×5 is *worse* than 3×3 for both filter families in both noise cases. A larger window averages or ranks over a larger neighbourhood, so it suppresses more noise but destroys more structure, and past a point the structural loss dominates.
3. **Failure mode.** Gaussian filtering fails *gracefully*: everything gets softer. Median filtering fails *characteristically*: fine texture is erased into flat patches and thin lines can be deleted outright, because a feature thinner than half the window is by definition the minority in it.
4. **Cost.** Convolution is a fixed multiply-accumulate per pixel and is separable; the median needs a sort per pixel and is not separable.

> **AMARAVEL BABU ARRAN's comment:** The practical rule I take from this is that the filter should match the *statistics* of the corruption, not the amount of it. Both noise types here are of similar visual severity, yet the correct choice of filter differs and choosing wrongly costs around 2 dB.

2.5. Suppressing Noise Interference Patterns

(b) The power spectrum

```
F = fft2(double(P));
S = abs(F).^2;                        % F is complex, S is real
imagesc(fftshift(S.^0.1)); colormap('default');
```

(c) Peak locations

Measured on the **unshifted** transform:

Peak	Index (MATLAB, 1-based)	Share of non-DC energy
1	**(17, 249)**	**20.75 %**
2 (conjugate twin)	**(241, 9)**	**20.75 %**

The next strongest local maximum anywhere in the spectrum holds 0.0740 %, so in power the interference peaks stand clear of everything else by a factor of **280**.

> **AMARAVEL BABU ARRAN's comment:** The two peaks are not independent. For a real-valued image the transform is conjugate-symmetric, F(−u,−v) = F*(u,v), which is why they are equal in magnitude and placed symmetrically about DC. This has a practical consequence for step (d): both must be zeroed*. Removing only one would leave an unmatched complex component, and the inverse transform would no longer be real: `ifft2` would return an image with a spurious imaginary part.

(d), (e) Notching and the inverse transform

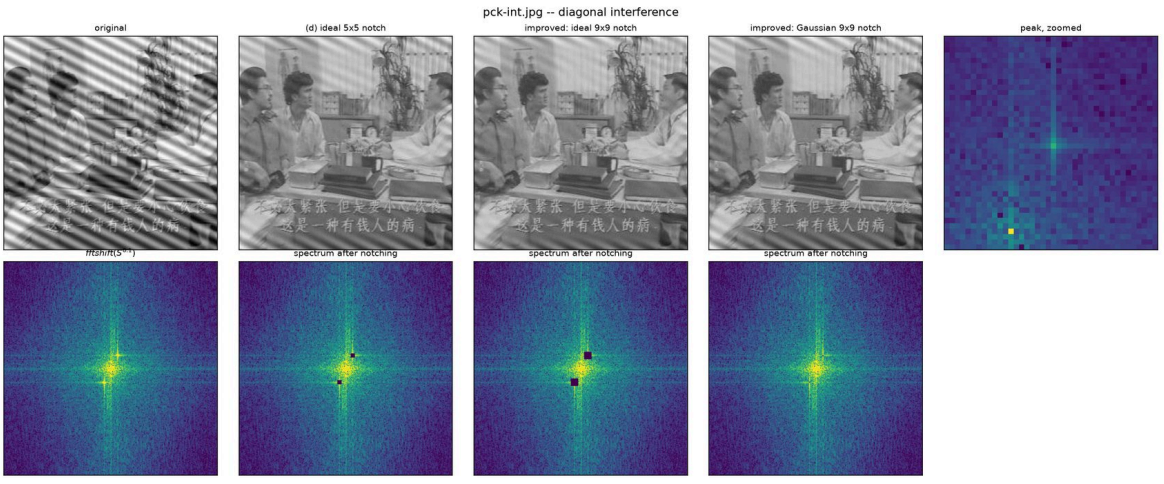


Figure 11. Notch filtering of pck-int.jpg. Top row: the original and the results of three notch designs. Bottom row: the corresponding power spectra, displayed as fftshift(S^0.1), with the removed neighbourhoods visible as dark squares either side of the centre.

The 5×5 ideal notch removes 57.1 % of the non-DC energy and drops the dominant component's prominence from **89.5 to 10.1**. The diagonal stripes are essentially eliminated and the underlying scene, two men in a room with subtitles, becomes legible.

How this relates to (c): directly. The 20.75 % energy share measured in (c) is a prediction that the interference is a single dominant sinusoid, and the result confirms it. Removing 50 coefficients out of 65 536, under 0.08 % of the transform, repairs the whole image, because in the frequency domain the corruption is concentrated rather than spread.

Attempting to improve the result:

Variant	Energy removed	Strongest component left
(d) ideal 5×5	57.12 %	10.1
ideal 9×9	58.99 %	9.1
Gaussian-profiled 9×9	57.54 %	**7.7**

The Gaussian-profiled notch is best, and notably it achieves this while removing *less* total energy than the wider ideal notch.

> **AMARAVEL BABU ARRAN's comment:** This is the frequency-domain dual of the same principle met in §2.3. An ideal notch is a rectangle in the frequency domain, and the inverse transform of a rectangle is a sinc, which rings. So a hard-edged stop band buys suppression at the cost of ripple around every edge in the image. Tapering the notch removes the discontinuity and so removes the ringing. My §2.8 program measures this directly on synthetic data where a clean reference exists, and finds the soft notch ahead by 5.6 to 13.4 dB at every radius tested.

Some residual banding remains. It is visible along the image axes in the spectrum, and it is caused by the image not being periodic: the DFT implicitly wraps the image, so the discontinuity between opposite edges leaks energy along the horizontal and vertical axes. Windowing the image before transforming would reduce it.

(f) Freeing the caged primate

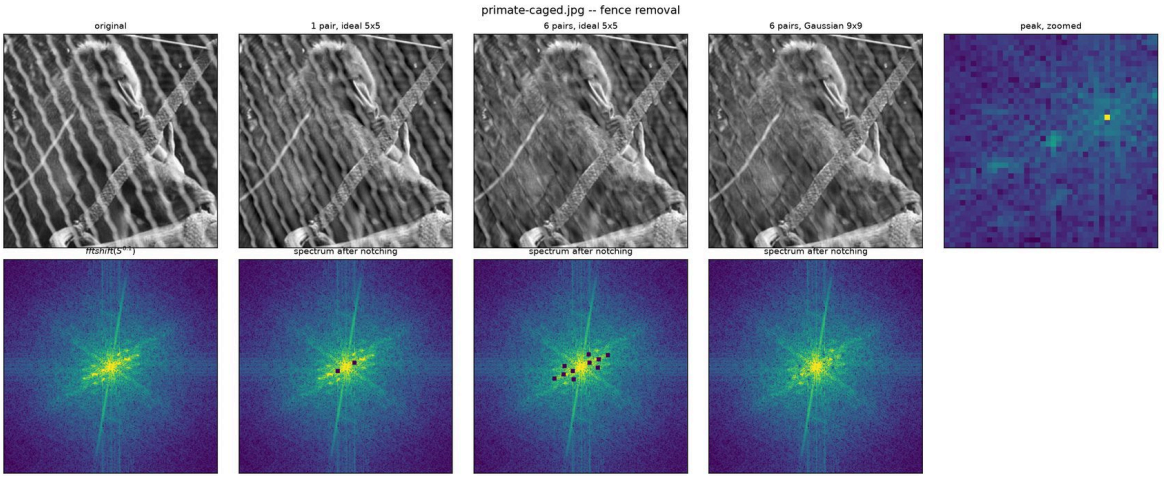


Figure 12. Point notching of primate-caged.jpg. Removing one conjugate pair barely affects the fence; five pairs are needed before it visibly weakens, because a fence contains a fundamental and its harmonics rather than a single frequency.

The fence fundamental holds only **5.0 %** of the non-DC energy, against 20.75 % for `pck-int.jpg`, and the detector finds a **harmonic series** rather than a single pair:

Pairs notched	Energy removed	Strongest left
original	n/a	17.6
1 pair	29.5 %	13.2
5 pairs	41.4 %	6.8

One notch pair barely helps. This is because a fence is not a sinusoid: it is closer to a square wave, and a square wave is a fundamental **plus its odd harmonics**, each of which needs its own notch.

Attempting to improve further. The spectrum shows the real obstacle: the fence's energy does not sit in isolated points at all, but along an extended **ridge** through the origin, found at 155°. A wavy fence has a slowly varying spatial frequency, which smears each harmonic into a continuous streak. Point notching cannot clear a ridge. So I notched along the ridge itself:

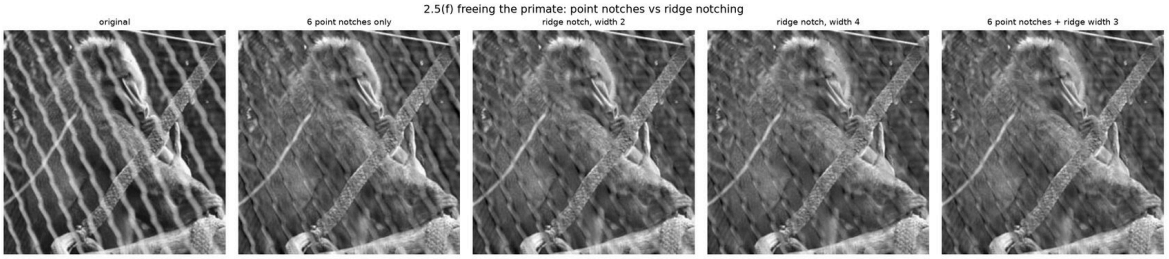


Figure 13. Improving on point notching by also attenuating the spectral ridge at 155 degrees along which the fence's energy lies. The combined notch gives the clearest view of the animal, while the diagonal strap, being non-periodic, is correctly left intact.

Variant	Energy removed
5 point notches only	43.57 %
ridge notch, width 2	44.59 %
5 point notches + ridge width 3	**51.57 %**

The combined notch gives the cleanest result, with the baboon's face and fur clearly legible, while the diagonal strap, which is a real object and not periodic, correctly survives.

> **AMARAVEL BABU ARRAN's comment:** A clean result is not achievable here, and the reason is fundamental rather than a matter of effort. The fence's ridge passes through the same frequency band the primate's own fur texture occupies. Any filter wide enough to remove the fence necessarily removes image content with it; the two signals are not separable in this domain. That is precisely why `pck-int.jpg` can be repaired almost perfectly and this image cannot. There the interference was a single narrow spike far from the image's own content, here it overlaps. I also noticed my prominence metric stops tracking visual quality once a wide band is removed, because removing the band changes the local background the prominence is measured against.

2.6. Undoing Perspective Distortion

(b) Corner correspondences

The manual obtains these with `ginput`, which is interactive. To keep the script reproducible I measured them once by segmenting the bright cover, fitting a straight line to each of its four edges, and

intersecting adjacent lines. The edge fits have sub-pixel residuals (0.31–0.94 px RMS), confirming the cover edges really are straight and the corners are well determined. The `ginput` call is retained in `Lab1.m` as a comment.

Corner	Image (X, Y)	Desired (x, y) mm
top-left	(142.55, 18.65)	(0, 0)
top-right	(307.74, 46.71)	(210, 0)
bottom-right	(255.37, 217.37)	(210, 297)
bottom-left	(4.25, 158.64)	(0, 297)

(c) The projective matrix

```
A = []; v = [];
for i = 1:4
    A = [A; X(i) Y(i) 1 0 0 0 -x(i)*X(i) -x(i)*Y(i); ...
            0 0 0 X(i) Y(i) 1 -y(i)*X(i) -y(i)*Y(i)];
    v = [v; x(i); y(i)];
end
u = A \ v;
U = reshape([u; 1], 3, 3)';
```

```
        [  1.194215e+00   1.179798e+00  -1.922385e+02 ]
  U  =  [ -4.939472e-01   2.907881e+00   1.618020e+01 ]
        [ -2.604151e-04   3.793097e-03   1.000000e+00 ]
```

Does the transformation give back the four corners? Yes, exactly:

Transformed	Target	Error
(−0.000000, 0.000000)	(0, 0)	2.9×10^{-14}
(210.000000, 0.000000)	(210, 0)	9.7×10^{-15}
(210.000000, 297.000000)	(210, 297)	5.7×10^{-14}
(0.000000, 297.000000)	(0, 297)	0

Maximum error **5.7×10^{-14} mm**, which is machine precision.

> **AMARAVEL BABU ARRAN's comment:** This verification is necessary but much weaker than it looks, and it would be a mistake to read it as proof the warp is good. A projective transform has 8 degrees of freedom and four correspondences supply exactly 8 equations, so the system is square and the solution reproduces its own input corners *by construction*: it would do so even if I had clicked the corners completely wrongly. All this checks is that the matrix was assembled and inverted correctly. Genuinely testing the warp requires evidence from points that were **not** used to fit it, which is what I use the straightness of the text baselines for below. Supplying more than four correspondences and solving by least squares would make the residual informative, since it would then have to compromise between them.

(d), (e) The warp

```
T  = maketform('projective', U');
P2 = imtransform(P, T, 'XData', [0 210], 'YData', [0 297]);
```

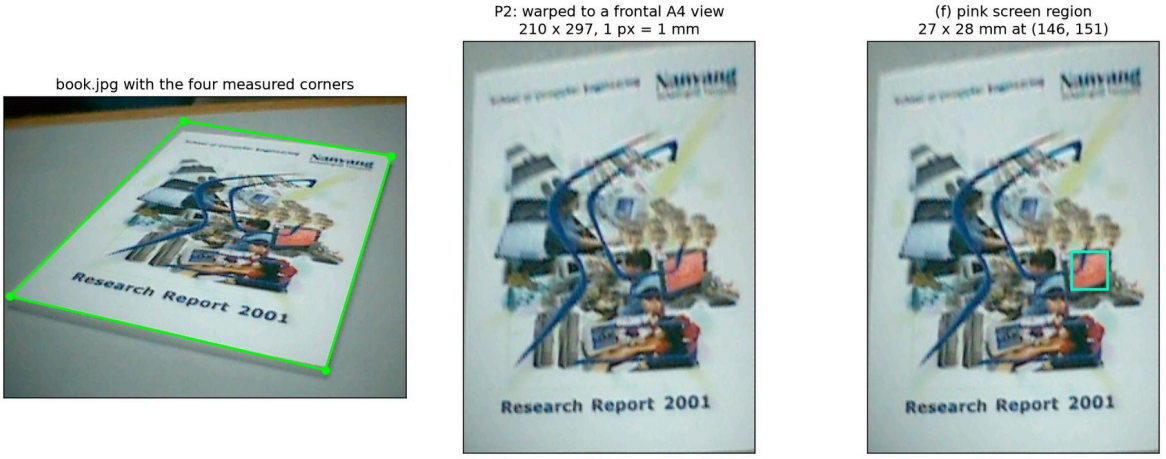


Figure 14. Perspective correction of book.jpg. Left: the slanted original with the four measured cover corners. Centre: the rectified A4 view at 1 pixel per millimetre. Right: the pink screen region located within the rectified page.

Is this what I expected? Yes. The cover is rectified to a frontal view: "Research Report 2001" now sits horizontal, the Nanyang logo is upright, and the page fills exactly 210 × 297 pixels at 1 pixel per millimetre.

Comment on quality, and reasons. The result is geometrically correct but noticeably soft, and the softness is *not uniform*:

1. **Non-uniform resolution, the dominant effect.** In the original, the book's near edge spans about 258 px for 210 mm (1.23 px/mm) while the far edge spans about 168 px for the same 210 mm (0.80 px/mm). The top of the page is therefore reconstructed from barely half the samples available at the bottom. Warping cannot create detail that was never captured, so the top of the rectified image is visibly blurrier, clearest in the "School of Computer Engineering" line, which is markedly less sharp than "Research Report 2001".
2. **Interpolation loss.** Output pixels map to non-integer source positions, so every value is a bilinear blend of four neighbours, itself a mild low-pass filter.
3. **Corner precision.** A one-pixel error in a corner propagates across the whole page; at the far edge one pixel is already 1.25 mm.
4. **The planarity assumption.** The model assumes a perfect plane. A paperback cover bows slightly, and that curvature cannot be represented by any 3×3 projective matrix: it is a systematic model error, not noise.
5. **Lens distortion.** Radial distortion makes straight lines slightly curved; the projective model has no term for it.
6. **Source resolution.** The original is only 320 × 240, so the cover occupies fewer pixels than the 210 × 297 output requests, so the warp is upsampling.

> **AMARAVEL BABU ARRAN's comment:** Points 4 and 5 are *model* errors and 1–3 are *sampling* errors, and the distinction matters for how you would fix them. More correspondence points and least-squares fitting would reduce 3 but do nothing for 4. Only a richer model, undistorting first or allowing a non-planar surface, addresses 4 and 5. The independent check I trust most is that the text baselines come out straight and parallel across the full page width: those points were never used in the fit, so their straightness is real evidence that the recovered homography is right.

(f) The pink screen region

Located by colour thresholding (red clearly dominant over both green and blue, adequate saturation and brightness), followed by morphological opening and closing and selection of the largest connected component:

27 × 28 mm, with its top-left corner at (146, 151) mm on the page.

This falls midway down the cover, between the Nanyang logo at the top and "2001" at the bottom, which is exactly where the manual describes it. Because the warp is scaled 1 px = 1 mm, the bounding box is read directly in millimetres with no further conversion, a useful side-effect of the chosen output scaling.

2.7. Two Perceptrons

Both algorithms are implemented as specified in the lecture slides and run on the slides' own example: $\mathbf{x(1) = [3, 3, 1]^T} \in \mathbf{c}_1$, $\mathbf{x(2) = [1, 1, 1]^T} \in \mathbf{c}_2$, $\alpha = 1$, $w(1) = [0, 0, 0]^T$. The bias is folded into $\mathbf{w}$ by the trailing 1.

(a) Algorithm 1: sequential sample update

```
a = X(i, :) * w;
if cls(i) == 1 && a <= 0
    w = w + alpha * X(i, :)';
elseif cls(i) == 2 && a >= 0
    w = w - alpha * X(i, :)';
end                                  % otherwise w is unchanged
```

Full trace:

k	sample	$w^T x$	rule	w after
1	x(1)	0.0	1	[3, 3, 1]
2	x(2)	7.0	2	[2, 2, 0]
3	x(1)	12.0	3	[2, 2, 0]
4	x(2)	4.0	2	[1, 1, −1]
5	x(1)	5.0	3	[1, 1, −1]
6	x(2)	1.0	2	[0, 0, −2]
7	x(1)	−2.0	1	[3, 3, −1]
8	x(2)	5.0	2	[2, 2, −2]
9	x(1)	10.0	3	[2, 2, −2]
10	x(2)	2.0	2	[1, 1, −3]
11	x(1)	3.0	3	[1, 1, −3]
12	x(2)	−1.0	3	[1, 1, −3]

Converges at k = 12 with $\mathbf{w} = [1, 1, -3]^T$, boundary $x_1 + x_2 - 3 = 0$, reproducing the lecture slides exactly, step for step, including every intermediate weight vector. 6 epochs, 7 updates.

(b) Algorithm 2: batch delta (LMS) rule

```
e    = r - X * w;                    % r = +1 for c1, -1 for c2
cost = 0.5 * sum(e .^ 2);
grad = -(X' * e);                    % accumulate over the whole epoch
w    = w - alpha * grad;
```

Converges to $\mathbf{w} = [0.5, 0.5, -2]^T$, boundary $x_1 + x_2 - 4 = 0$.

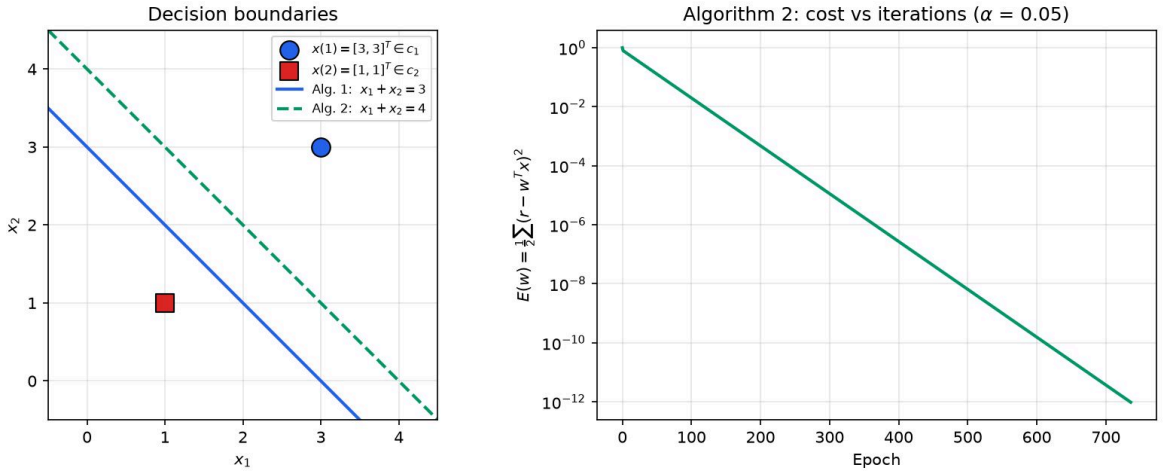


Figure 15. Comparison of the two perceptrons. Left: Algorithm 1's boundary passes close to x(2), while Algorithm 2's bisects the two samples exactly. Right: the cost of Algorithm 2 decaying geometrically over its 737 epochs.

(c) Comparison and comments

1. The learning rate is not transferable. Algorithm 2 **diverges** at the α = 1 that Algorithm 1 uses happily.

> **AMARAVEL BABU ARRAN's comment:** This is not a tuning accident, it is predictable in closed form. The batch update is $w \leftarrow (I - \alpha R)w + \alpha p$ with $R = \Sigma_i x_i x_i^T$, a linear iteration which converges only when every eigenvalue of $(I - \alpha R)$ lies inside the unit circle, i.e. $\alpha < 2/\lambda_max(R)$. Here $\lambda_max = 21.63$, giving $\boldsymbol{\alpha < 0.0925}$. At $\alpha = 1$ the iteration matrix has spectral radius ≈ 20 and the weights blow up geometrically. At $\alpha = 0.05$ it converges in 737 epochs. Algorithm 1 has no such constraint because it does not descend a cost surface: with $\alpha = 1$ it merely rescales $\mathbf{w}$, and the sign of w^Tx is invariant to positive scaling, so the learning rate cannot destabilise it at all.

2. The boundaries differ, and Algorithm 2's is symmetric. Perpendicular distances from the boundary to the two training points:

	Boundary	to x(1)	to x(2)	Margin ratio
Algorithm 1	$x_1 + x_2 = 3$	+2.1213	−0.7071	**0.333**
Algorithm 2	$x_1 + x_2 = 4$	+1.4142	−1.4142	**1.000**

Algorithm 2's line bisects the two points. Algorithm 1's sits three times closer to x(2) than to x(1).

The bisection is exact analytically, not merely close. With three unknowns and two constraints, gradient descent from $\mathbf{w} = 0$ converges to the minimum-norm solution of $w^Tx(1) = +1$ and $w^Tx(2) = -1$. Subtracting those gives $w_1 + w_2 = 1$ and $w_3 = -2$, and the minimum-norm choice splits the first evenly, so $\mathbf{w} = [0.5, 0.5, -2]^T$ exactly and the margin ratio is exactly 1. The run stops at a cost tolerance of 10^{-12} rather than at the limit, so the iterate it actually reports is $[0.5, 0.5, -1.999998]^T$ with a ratio of 0.9999992, which is the analytical answer to six decimal places.

> **AMARAVEL BABU ARRAN's comment:** The mechanism is the stopping condition. Algorithm 1 halts the moment both samples are classified correctly. It is satisfied by *any* separating line and stops at whichever one it happens to have reached, so its answer depends on the order the samples were presented and on the initial weights. Algorithm 2 never stops at "good enough": it minimises $\Sigma(r - w^Tx)^2$, which has a unique minimiser, so it converges to the same place regardless of ordering. The symmetry is a direct consequence of the targets being ± 1: forcing $w^Tx(1) = +1$ and $w^Tx(2) = -1$ places the $w^Tx = 0$ contour precisely halfway between them. This confirms the slides' claim quantitatively: the margin ratio is 1.000, not merely "smoother".

3. Convergence behaviour. Algorithm 1 converges in 12 sample presentations with 7 discrete jumps; Algorithm 2 takes 737 epochs of small smooth steps, its cost decaying geometrically.

4. Which is better? For this separable two-point problem Algorithm 1 is far cheaper. But Algorithm 2's boundary is the one I would prefer: a boundary that bisects the classes generalises better to unseen data than one that grazes a training point, and the same squared-error formulation still converges when the data are *not* linearly separable, where Algorithm 1 never terminates at all.

2.8. Using Your Creativity

Program: `notch_filter_studio.html`, which opens in any browser. No install, no libraries, no plugins, and it works offline.

(a) Motivation and problem definition

§2.5 removes interference by transforming to the frequency domain, zeroing two measured coordinates, and transforming back. Run as a script, the interesting part is invisible: you read two numbers off an axis, hard-code them, and look at the output. The questions that decide whether the method actually works are never confronted: how do you know *which* peaks are interference; why must peaks be removed in pairs; how large should the notch be and what does enlarging it cost; and why can a fence not be removed as cleanly as diagonal stripes?

Problem: build a tool that makes the forward transform, the surgical edit and the inverse transform directly manipulable, so each question is answered by measurement rather than assertion.

(b) How the program was developed

A single HTML file with **zero external dependencies**, deliberately stricter than §2.8(d) requires, and stricter than the reference example, which loads three.js from a CDN and so fails without a network. Every numerical routine is written from scratch: a radix-2 Cooley–Tukey FFT with bit-reversal permutation, applied separably; the §2.5(b) display convention `fftshift(S.^0.1)`, percentile-clipped because min–max normalisation washed it out; click-to-notch with coordinates mapped back by u = (cx + N/2) mod N and every notch automatically mirrored to its conjugate twin.

The peak detector took three attempts, and the failures were the instructive part:

1. **Fixed count (take the 6 strongest).** Wrong by construction. Forcing the detector to a fixed number on the single-sinusoid case gives 51.32 dB at **1** pair, 31.61 dB at 2 and 28.53 dB at 3 or more: a penalty of nearly 23 dB for notching components that were never there. The fence meanwhile genuinely contains **3**. The right number is a property of the image, not a constant.
2. **Threshold at a multiple of the global median magnitude.** Also wrong: an image spectrum spans orders of magnitude, so ordinary structure clears any global cut. Run without a prominence test on a *clean* image it still finds "peaks" and destroys what it touches, giving 37.66 dB at one pair and 25.59 dB by eight. The shipped detector finds none and leaves the image untouched.
3. **Local prominence plus energy share**: the final design. A candidate must be a 5×5 local maximum, exceed its own surrounding annulus by a set factor, and carry ≥ 0.1 % of the non-DC energy. Prominence rejects image structure, which sits on smooth ridges; the energy test rejects the Nyquist corner and the "most prominent peak" that exists in every image even with no interference present.

> **AMARAVEL BABU ARRAN's comment:** One further bug is worth recording because it was a modelling error rather than a coding one. My first synthetic test pattern contained seven fine lines evenly spaced 4 px apart. Evenly spaced lines *are* a periodic signal, so they planted genuine spikes at v ≈ 256/4 = 64 in the supposedly clean reference image, spikes no detector could possibly distinguish from injected interference. I replaced the detail with a deterministic pseudo-random texture, which is broadband. The lesson transferred directly to §2.5(f): the reason the fence resists removal is the same reason that test pattern broke my detector: periodic *content* and periodic *corruption* are indistinguishable in the frequency domain.

(c) Results

Source	Pairs found	PSNR before	PSNR after
Clean test pattern	**0**	∞	∞ (untouched)
Diagonal sinusoid (as `pck-int.jpg`)	**1**	18.29 dB	**51.32 dB**
Periodic fence (as `primate-caged.jpg`)	**3**	19.01 dB	**32.95 dB**

The detector recovers exactly the right number of components in every case: none when there is no interference, one pair for a single sinusoid, and three for the fence, being its fundamental plus the 3rd and 5th harmonics, which is precisely what a near-square-wave occluder contains.

Ideal versus Gaussian notch, sweeping the radius on the diagonal case:

Radius	Ideal (hard zero)	Gaussian (soft)	Gaussian advantage
1	55.50 dB	**68.87 dB**	13.37 dB
2	51.32 dB	**57.56 dB**	6.24 dB
3	47.47 dB	**53.40 dB**	5.93 dB
5	42.45 dB	**48.28 dB**	5.83 dB
8	38.12 dB	**43.76 dB**	5.64 dB

The soft notch wins at every radius, by between 5.6 and 13.4 dB, and both degrade steadily as the notch widens. This is §2.5(e)'s "attempt to further improve the result", quantified.

> **AMARAVEL BABU ARRAN's comment:** A third result emerged unplanned while tuning the demo. Moving the interference to a higher frequency, further from the image's own spectral content, raised the achievable PSNR from 38.5 dB to **51.3 dB with no change whatsoever to the algorithm**. How well interference can be removed is therefore not a property of the filter at all, but of how well separated the interference is from the signal. That is the honest explanation for the whole of §2.5: `pck-int.jpg` is repairable almost perfectly because its interference is a narrow spike far from the image's content, and `primate-caged.jpg` is not, because the fence's harmonics lie inside the band the fur texture occupies.

(d) The program

`creativity/notch_filter_studio.html`, submitted alongside this report. Immediately executable: open it in a browser.

A. Files Submitted

File	Contents
`Lab1.m`	Complete source for all seven sections, in one file (req. 3b)
`notch_filter_studio.html`	The §2.8 program (req. 3c)
`report.pdf`	This report (req. 3a)